

LEVEL 0: NOMINAL CBF State: Closed-Loop Control 1000 Hz Minimal-Intervention QP	Activation Trigger: Safe set invariant satisfied: $h(x) \geq 0$ Stopping margin clear: $d_{\text{clear}} > d_{\text{stop}}(v)$ Valid unexpired intent lease: $\text{TTL} > 0$	Hardware & Actuator Consequence: Permitted setpoint: $u_t = \Pi_{\mathcal{U}_{\text{safe}}}(p_t)$ 3-phase inverter gate drivers actively PWM modulated Full task execution authority
LEVEL 1: CATEGORY 2 STOP Standard: IEC 60204-1 Cat 2 Controlled Position / Coordinate Hold	Activation Trigger: MPU Intent lease expired: $t > t_{\text{expire}}$ IPC heartbeat missing: $15 \text{ ms} < \Delta t \leq 50 \text{ ms}$ Non-critical sensor degraded / occluded	Hardware & Actuator Consequence: Actuator coils remain energized Closed-loop PID holds coordinate $x(t) = x_{\text{hold}}$ System awaits fresh valid MPU intent lease
LEVEL 2: CATEGORY 1 STOP Standard: IEC 60204-1 Cat 1 (SS1) Safe Stop 1: Controlled Deceleration	Activation Trigger: Stopping clearance breach: $d_{\text{clear}} \leq d_{\text{stop}}(v)$ Safe set boundary violated: $h(x) < 0$ Joint kinematic / velocity envelope exceeded	Hardware & Actuator Consequence: Active maximum dynamic braking: $a_{\text{brake}} = a_{\text{max}}$ Mechanism brought to zero velocity ($v \rightarrow 0$) Actuator power removed after standstill is reached
LEVEL 3: CATEGORY 0 STOP Standard: IEC 60204-1 Cat 0 (STO) Safe Torque Off / Direct Power Cut	Activation Trigger: Physical E-Stop button pressed Hardware watchdog timer trip: $\Delta t > 50 \text{ ms}$ Hardware over-current trip: $I_{\text{phase}} > I_{\text{trip}}$	Hardware & Actuator Consequence: Hardware optocoupler pulls EN_GATE low in $< 5 \mu\text{s}$ Zero software dependency; gate drive power isolated Spring-actuated mechanical friction brakes engage

INCREASING
SEVERITY
& HARDWARE
ISOLATION

The Asynchronous State Re-Anchoring Handshake Upon Safety Intervention

1. MCU Intercepts & Latches:

Safety enforcer vetoes trajectory p_t

MCU asserts atomic flag `IS_INTERVENTION = 1`

Latches resting coordinate x_{veto} in SRAM

2. IPC Mailbox Notification:

Hardware interrupt posted to Linux MPU

Telemetry stream tagged with intervention ID

Prevents DAGger covariate policy corruption

3. MPU Pipeline Flush & Reset:

MPU aborts obsolete action chunk buffer

Flushes transformer KV-cache

Re-seeds diffusion planner from true x_{veto}